Complete manuscript comparison

Overview v17: locally toroidal extensions

This edition covers 19 topics and 51 measured originals. The 106 accessible originals comprise 51 measured mathematical originals, 54 awaiting mathematical recovery and measurement, and one editorial, L022. The 55 unavailable originals remain archived. L022 contains no principal mathematical result to recover: it stays accessible, with no factor or measured credit.

L121 joins A. All 18 other whole-topic records and all 50 earlier individual records retain their exact v16 values. A's whole factor changes from 1.395290 to 1.354347; the new L121 conditional is 1.071157. The complete written factor for each topic is $\sum_i c(S_i)/(c(U) + \sum_i c(A_i))$: whole common theory and all complete applications are paid once within that topic.

Each topic retains its adopted nonwhitespace-character coordinate. There is no pooled factor or cross-topic deduction, and no inferred minimum proof length. Executable software is excluded on both sides; mathematical proof and algorithm prose remain counted.

Topic	Originals	Sources	Common	Applications	Factor
A	9	273,937	66,140	136,125	1.354
B	5	124,906	44,905	69,265	1.094
C	5	118,433	30,437	61,248	1.292
D	3	47,894	15,632	29,855	1.053
E	5	140,286	57,473	32,337	1.562
F	2	69,825	15,876	27,398	1.614
G	2	73,199	30,928	85,207	0.630
Representation	2	60,472	36,715	45,437	0.736
Chamber	2	34,177	15,199	21,254	0.938
Matching	1	37,951	16,732	17,168	1.119
ChiralEnumeration	1	27,594	8,831	23,171	0.862
Operations	3	200,630	58,215	100,379	1.265
Gluing	1	27,081	28,416	11,565	0.677
RankContraction	1	15,530	6,937	10,555	0.888
LocallyProjective	2	26,490	25,498	11,966	0.707
MathieuCensus	1	13,736	5,516	35,862	0.332
RankAugmentation	1	33,604	13,904	14,096	1.200
Mixes	3	156,570	34,757	68,693	1.513
LCD	2	58,756	32,300	66,886	0.592

The 19-topic CSV and 51-original CSV preserve exact fractions and complete component costs. Factors below one mean written expansion and remain visible. The separate source admissions define the corrected principal contracts; this overview creates no additional proof, question or formalization credit.

Individual comparisons: first 26 originals

The conditional factor is $c(S_i)/c(A_i)$ after acquiring the *whole* common theory and all actual required mathematical, computational and data premises. Every complete local application or companion bundle remains charged.

Original	Topic	Source	Application	Factor
L032	A	16,558	8,228	2.012
L034	A	21,433	5,801	3.695
F016	A	43,983	15,436	2.849
F013	A	25,562	6,241	4.096
L035	A	51,019	14,514	3.515
N009	A	36,824	28,130	1.309
L128	A	18,424	8,825	2.088
L012	A	32,767	23,401	1.400
L121	A	27,367	25,549	1.071
N001	B	14,519	7,371	1.970
N003	B	17,780	9,155	1.942
N008	B	18,009	11,132	1.618
L030	B	22,704	12,286	1.848
L040	B	51,894	29,321	1.770
L088	C	13,166	11,311	1.164
N013	C	27,361	15,815	1.730
L108	C	12,849	10,075	1.275
F014	C	20,577	6,962	2.956
L068	C	44,480	17,085	2.603
L066	D	20,916	10,989	1.903
L074	D	20,661	13,367	1.546
L104	D	6,317	5,499	1.149
L079	E	21,212	1,858	11.417
L083	E	20,904	2,922	7.154
L021	E	29,594	14,452	2.048
L056	E	15,871	4,144	3.830

A pays the same entire 23-page common paper and all nine applications, now 56 pages, against nine complete originals totaling 131 pages:

$$R_A = \frac{273,937}{66,140 + 136,125} = 1.354347.$$

L121 contributes its complete 12-page source and 11-page application: $27,367/25,549 = 1.071157$ conditionally. It uses no sibling application as a premise. The earlier eight A conditional fractions are unchanged in all thirty coordinates.

Individual comparisons: remaining 25 originals

Original	Topic	Source	Application	Factor
L043	E	52,705	8,961	5.882
L014	F	25,574	11,504	2.223
L007	F	44,251	15,894	2.784
L009	G	49,891	23,544	2.119
L092	G	23,308	61,663	0.378
F010	Representation	25,348	10,985	2.308
L126	Representation	35,124	34,452	1.020
L008	Chamber	20,159	7,571	2.663
N016	Chamber	14,018	13,683	1.024
F015	Matching	37,951	17,168	2.211
L028	ChiralEnumeration	27,594	23,171	1.191
N005	Operations	122,002	49,595	2.460
N006	Operations	55,740	38,479	1.449
N007	Operations	22,888	12,305	1.860
L010	Gluing	27,081	11,565	2.342
L037	RankContraction	15,530	10,555	1.471
L084	LocallyProjective	2,071	4,857	0.426
L095	LocallyProjective	24,419	7,109	3.435
L073	MathieuCensus	13,736	35,862	0.383
L020	RankAugmentation	33,604	14,096	2.384
F005	Mixes	57,753	31,077	1.858
L016	Mixes	35,506	15,125	2.348
F007	Mixes	63,311	22,491	2.815
L050	LCD	24,674	37,432	0.659
F017	LCD	34,082	29,454	1.157

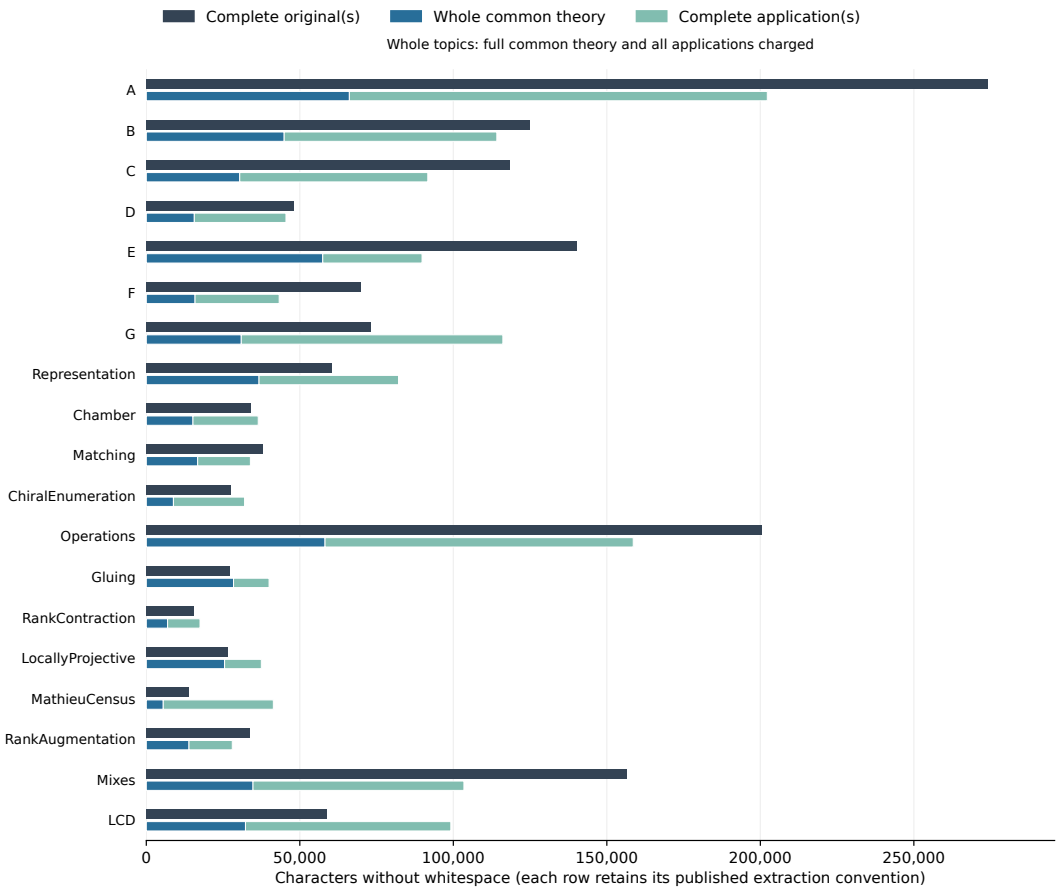
A's separate inventory contains 45 whole distinct files: 20 required programs/data, three auxiliaries, six operational runners, and 15 partial Lean modules plus one dependency manifest. The required files occupy 105,850 original bytes and 92,586 primary nonspace characters. They remain proof premises. The old 30-page printed appendices cover only eleven old required files; they do not represent the complete enlarged collection.

The six added Lean declarations verify specified finite relations and nonconsequences, not all group orders, infinite families or geometry. L121's corrected claims and retained Euclidean/toroidal inputs remain explicit. Its diagrams and L035's unextracted image labels are not fully priced.

All software-inclusive and pooled factors remain null. Historical software, external proofs, complete graphics and full formalization costs are unknown. Operations retains its text-only sensitivity; E retains the source's Magma/L_2-quotient attribution and all source qualifications. Prior overviews remain historical records.

Whole topics

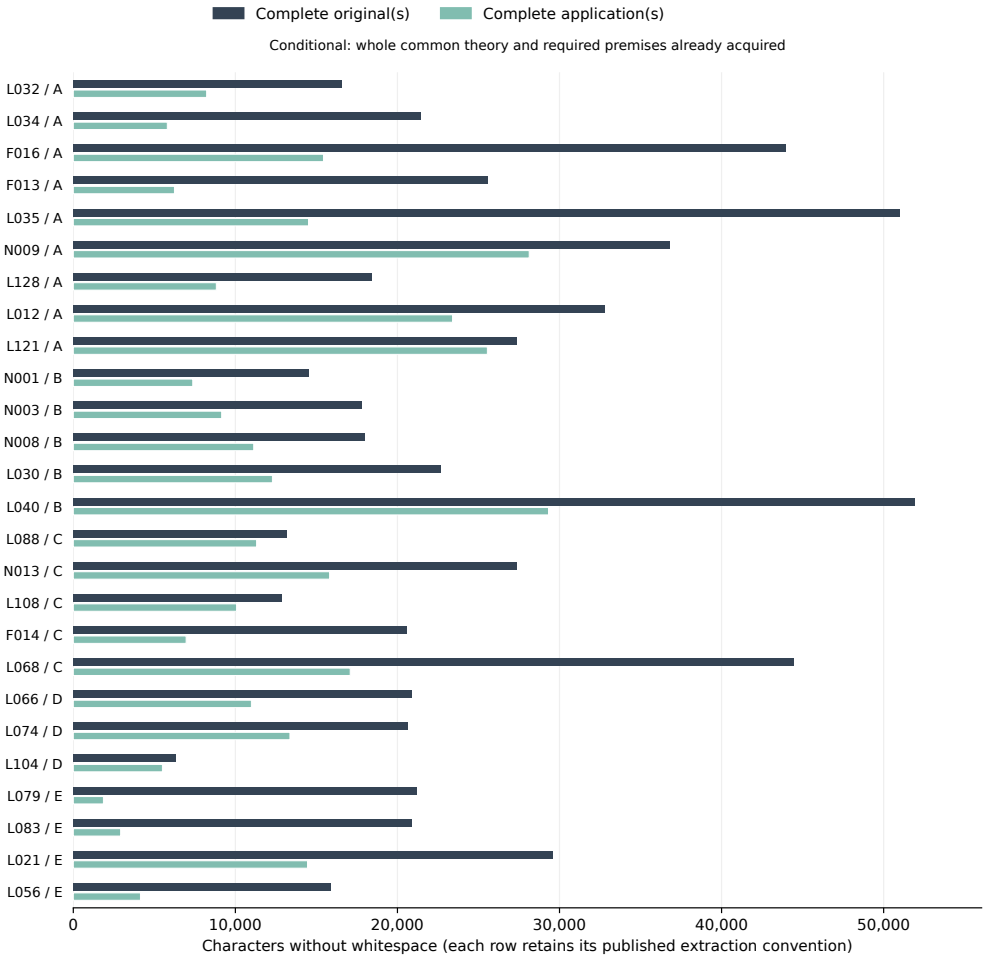
Grey bars show the complete originals in each row's adopted coordinate. Replacement bars stack the entire common theory and all applications. The axis starts at zero; the table gives the factors.



The 57 bars cover all 19 topics. Necessary finite certificates and retained external premises do not become optional because their software or external proof costs are outside this manuscript comparison.

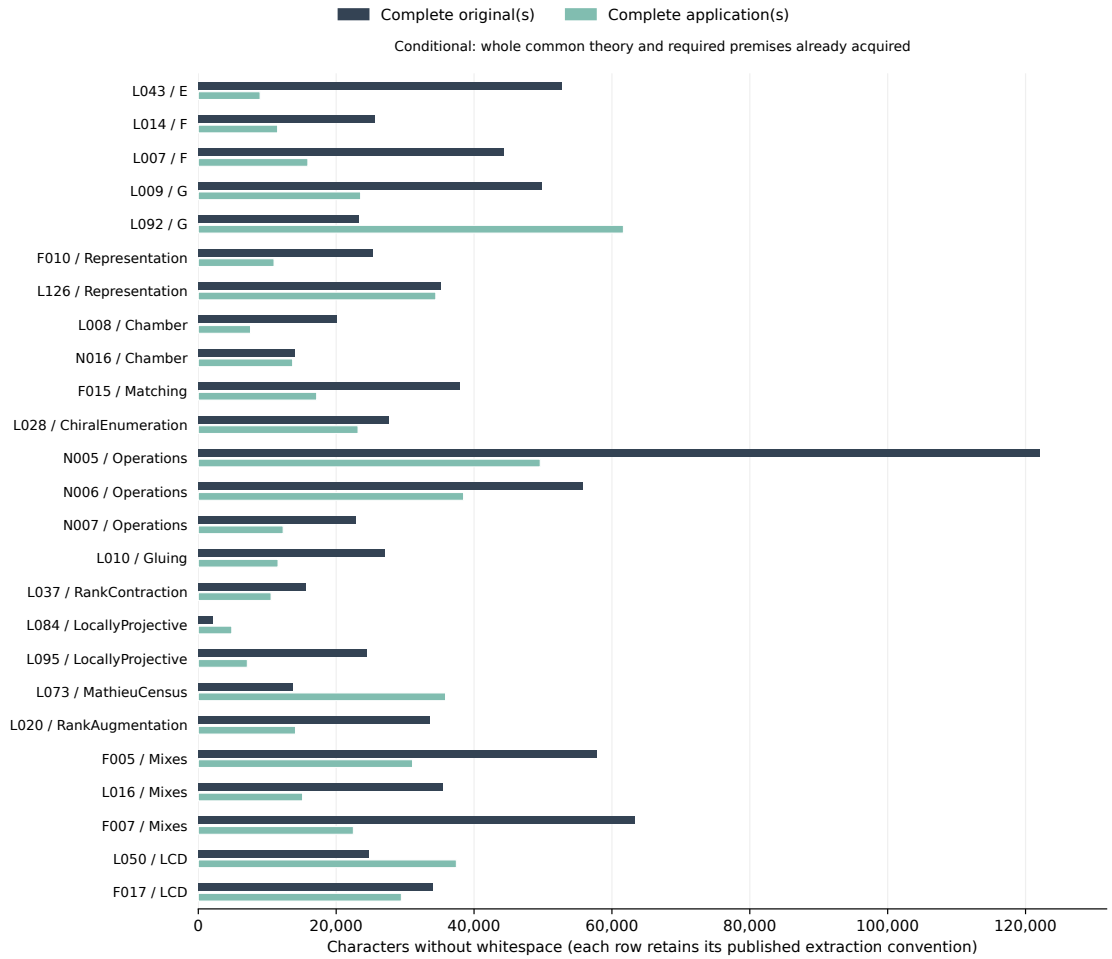
Conditional lengths: first 26 originals

Whole common theory and required premises are acquired in advance. Grey and teal bars compare complete source and local application text; the whole-topic table still pays the common theory.



Conditional lengths: remaining 25 originals

All complete companion papers remain charged. Source order is documentary, not an optimal acquisition order or a claimed sawtooth.



This edition supplies 70 exact manuscript comparisons and 159 plotted bars. The A9 numerical author is rewrite_termination; enumeration_volume performed its distinct numerical review and previously authored the inputs and A8 numerical mechanics. This overview reuses those accepted results with the prior roles disclosed. L022 has no mathematical recovery credit or numerical bar. No new conjecture or priority claim is made.